

# CrystalRL — Crystal Clear Reinforcement Learning

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## Abstract

Post-hoc explanation of a trading policy answers the wrong question: it tells a story *about* a black box while the box goes on being one. We report the construction of an agent that is legible *by* construction — and the measurement discipline that keeps “legible” from being a vibe. The paper traces one concrete journey: from **CHRL**, a constrained hierarchical PPO trader whose three decision layers are readable but whose core is a 64-dimensional unnamed latent with  $\sim 214$  tuning knobs, to **CRYSTAL-1**, whose memory is a *named* regime posterior  $b_t = \mathbb{P}(\text{bear} \mid \mathcal{F}_t)$  computed by an explicit Bayes filter, whose policy is a readable rule (a small decision tree) at full performance parity, and whose command surface is five named levers. All evidence is evaluated **on real market data**: on the six-asset US universe the readable dynamic policy beats the best static portfolio by +10.5pp goal probability; an LLM-driven improvement loop lifts the conservative investor’s goal probability from 1% to 52% at an unchanged risk budget (65 cycles, all 15 accepted changes confirmed on held-out data); the quoted 80% floors held in 5/5 US profiles across 576 point-in-time cells; and one certified readable rule — “ $b_t > 0.66 \Rightarrow \text{exposure } 0.74$ ” — ran on 629 untouched days at Sharpe 1.46 vs 1.39 buy-and-hold with max drawdown  $-12.9\%$  vs  $-15.5\%$ . On the identical scalar,  $\text{CRYSTALSCORE} = \text{Faithfulness} \times \text{Simulatability} \times \text{Stability}$ , the journey moves the agent from 0.15 to 0.92 on real data, and the entire gap is Simulatability. A separate section asks a question the real market cannot answer — *can one design a market that rewards only a clean mind?* — and uses a synthetic environment purely as instrument calibration, fenced off from every market claim by an explicit value-of-information gate. We close with twelve falsifiable hypotheses for the interpretability track.

## 1 Introduction: the question behind “explainable”

When a reinforcement-learning agent trades real money, “why did it do that?” is not a curiosity — it is a regulatory demand, a debugging requirement, and (in our companion work on self-improving systems) a safety precondition: *an agent that changes itself may only be allowed to do so to the degree it can be read*. The dominant practice answers with post-hoc explanation: train a black box, then probe it. Rudin [2019] names the flaw: for high-stakes decisions, do not explain the black box; build the model interpretable in the first place. The critique is well known. What is scarcer is a *worked example with a measurement discipline*: an agent built legible by construction, evaluated on real market data against a strong black-box predecessor, with the legibility difference expressed as a number both architectures can be scored on — and with the costs stated.

This paper is that worked example. It has the shape of a journey because we walked it in order: we first built the transparent-*layers* agent (CHRL, §2), measured where its transparency stops, then rebuilt the core so the stopping point disappears

(CRYSTAL-1, §3). Everything that counts as evidence is computed on real data (§4–§7); the one synthetic environment we built is used for instrument calibration only, behind an explicit gate (§8).

Contributions:

1. **A final CHRL→CRYSTAL-1 comparison on real data** (Table 2): architecture, control surface, behavior, and one shared legibility scalar.
2. **CrystalScore and its measurement protocol** (§5): faithfulness as intervention-following (Eq. 13), simulatability as a parity-constrained short story (Eq. 14), stability as seed replication — with the MDL deficit (Eq. 16) as the non-saturating companion.
3. **A fully specified data and metrics section** (§4): the US universe, its provenance and point-in-time discipline, and every evaluation formula.
4. **A taxonomy of behavioral influence with equations and real-data outcomes** (§6): reward shaping, teachers, belief interventions, architecture — including the intervention the safety bar *refused*.
5. **Real-data results** (§7): the +10.5 pp readable champion; the 1% → 52% per-profile improvement loop; 5/5 US profiles holding their quoted floors; a certified readable rule on untouched data; the transparency audit.
6. **The designed-market experiment** (§8): *can a market be built that rewards only a clean mind?* — with the VoI gate that keeps its results out of market claims.
7. **A think-big research program** (§11): twelve falsifiable hypotheses, each with a kill test.

## 2 The predecessor: CHRL, or how far “transparent layers” go

CHRL (Constrained Hierarchical RL) was our best answer to interpretability *within* the standard recipe. Decisions are factored into three readable layers — a risk layer choosing cash-vs-risky exposure on a ~20-day rhythm, a group layer routing capital through stock clusters, and a stock layer picking names weekly — wrapped in explicit guardrails: exposure pacing, confidence-scaled sizing, event triggers, top- $K$  routing, and a buy gate. Hierarchy as a partial legibility device has a long RL lineage [Sutton et al., 1999, Bacon et al., 2017]. The core is a PPO policy [Schulman et al., 2017] trained with the clipped surrogate

$$L^{\text{CLIP}}(\theta) = \mathbb{E}_t \left[ \min(\rho_t(\theta) \hat{A}_t, \text{clip}(\rho_t(\theta), 1-\epsilon, 1+\epsilon) \hat{A}_t) \right], \quad \rho_t(\theta) = \frac{\pi_\theta(a_t | s_t)}{\pi_{\theta_{\text{old}}}(a_t | s_t)}, \quad (1)$$

whose state  $s_t$  passes through a 64-dimensional *unnamed* latent.

It works as a trader: in four-fold walk-forward validation on the real US panel the champion variant delivered mean validation return 10.05%, Sharpe 1.09, max drawdown −7.34%, mean cash weight 41.24%, and 0.84%/day L1 turnover. Its *post-hoc* interpretability program also delivered: self-supervised primitive discovery over the frozen policy found behavioral motifs whose targeted edits improved a frozen real 2022–23 rollout without retraining (+0.54% return / +0.049 Sharpe for the

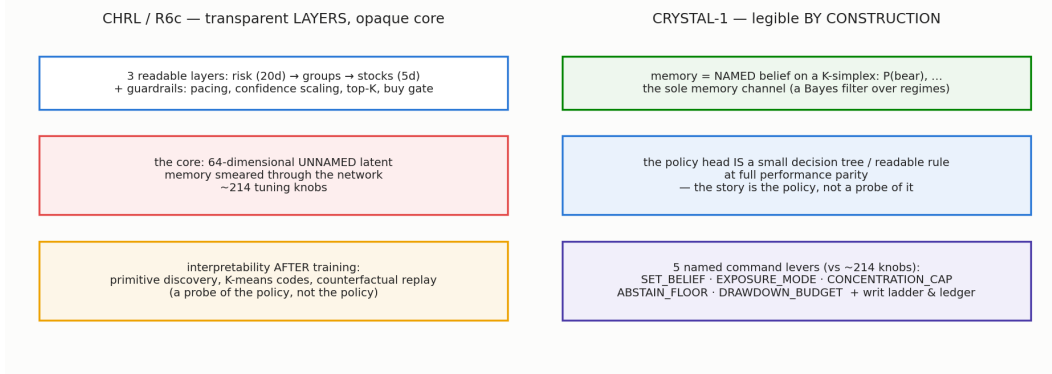


Figure 1: The journey in one picture. CHRL/R6c factors decisions into readable layers but keeps an unnamed 64-d latent core with  $\sim 214$  knobs, explained only after training. CRYSTAL-1 makes the explanation the mechanism: a named belief as the sole memory channel, a readable rule as the policy itself, and five named command levers.

best control-adjusted counterfactual). Post-hoc analysis is not nothing; we state this against our own thesis.

But every question we most cared about hit the same wall. Memory is smeared through the network, so no bound exists on what a state change can touch. The control surface is  $\sim 214$  engineering knobs, none a semantic command. Steering means forcing the latent toward a post-hoc K-means centroid and checking that behavior moves *consistently* — a correlation, not an intervention. And when we scored the deployed agent’s own stance log on the simulatability metric of §5, a  $K \leq 9$  story reproduced only 0.244 of its behavior: the latent is genuinely incompressible at a human-story budget. Transparent layers, opaque core (Figure 1).

### 3 CRYSTAL-1: legible by construction

CRYSTAL-1 rebuilds the core around three commitments.

**1. Memory is a named posterior.** The agent’s only recurrent state is a belief over  $K$  market regimes — a point on the simplex  $\Delta^{K-1}$  — in the POMDP sense of a belief state [Kaelbling et al., 1998], over Hamilton-style regimes [Hamilton, 1989]. For the  $K=2$  US machinery, with regime densities  $f_z(r) = \mathcal{N}(r; \mu_z, \sigma_z^2)$ ,  $z \in \{\text{bull}, \text{bear}\}$ , and a sticky transition prior  $\pi_{zz'}$ , the belief is the explicit Hamilton filter

$$b_t = \mathbb{P}(z_t = \text{bear} \mid \mathcal{F}_t) = \frac{f_{\text{bear}}(r_t) \tilde{b}_t}{f_{\text{bear}}(r_t) \tilde{b}_t + f_{\text{bull}}(r_t) (1 - \tilde{b}_t)}, \quad (2)$$

$$\tilde{b}_t = \pi_{BB} b_{t-1} + \pi_{GB} (1 - b_{t-1}),$$

where  $\mathcal{F}_t$  is the information available at  $t$  (§4). *Named* is load-bearing: because the coordinate has a meaning, a write to the belief is a sentence (“believe bear at 0.8”), and because  $b_t$  is the sole memory channel, the blast radius of any memory intervention is bounded by construction.

**2. The policy head is the story.** The decision layer is a small tree over  $[b_t, \text{inventory}, \text{time}]$ . In the soft form [Frosst and Hinton, 2017] each inner node  $i$

routes by a sigmoid gate and each leaf  $\ell$  holds a distribution over actions:

$$p_i(x) = \sigma(\beta(w_i^\top x + c_i)), \quad P^\ell(x) = \prod_{i \in \text{path}(\ell)} p_i(x)^{\mathbf{1}[\ell \prec i]} (1 - p_i(x))^{\mathbf{1}[i \searrow \ell]}, \quad (3)$$

$$\pi(a \mid x) = \sum_{\ell} P^\ell(x) Q_a^\ell,$$

with the inverse temperature  $\beta$  annealed so the trained tree is effectively hard. The constraint is that the tree *is* the policy — there is no residual network for behavior to hide in — and it must pay no performance tax (§7). At the limit the readable rule degenerates to a sentence; the certified US rule of §7.4 is literally

$$e_t = \begin{cases} 0.74, & b_t > 0.66 \\ 1.00, & \text{otherwise,} \end{cases} \quad (4)$$

where  $e_t$  is the equity exposure.

**3. Commands are few, named, and governed.** The  $\sim 214$ -knob surface collapses to **five agent-facing levers**:

SET\_BELIEF · EXPOSURE\_MODE · GROUP\_CONCENTRATION\_CAP  
ABSTAIN\_FLOOR · DRAWDOWN\_BUDGET

(the concentration cap is *hard*, fixing the predecessor’s soft-cap failure). Everything else is engineering-fenced. A write is the *do*-operator of Pearl [2009] made concrete:  $\text{SET\_BELIEF}(b^*, w)$  realizes

$$b_t \leftarrow (1 - w) b_t + w b^*, \quad w \in [0, 1], \quad (5)$$

and a *dose* nudge of  $\delta$  nats acts in log-odds,  $\text{logit}(b_t) \leftarrow \text{logit}(b_t) + \delta$ . Writes must pass a non-inferiority bar on the certified objective before they are allowed to persist (§6).

## 4 Data and evaluation metrics

Everything quantitative in §7 is computed on the following real data, under the following formulas. (The same universe and discipline underlie the companion paper *Hello Crystal*; this section makes the paper self-contained.)

### 4.1 The US universe and its provenance

Daily total-return series, 2000–2026 where listed, for six liquid US-listed assets plus cash:

| ticker | asset class                 | role in the menu   | listed since |
|--------|-----------------------------|--------------------|--------------|
| SPY    | US large-cap equities       | the growth engine  | 1993         |
| EFA    | developed ex-US equities    | diversifier        | 2001         |
| IEF    | 7–10y US Treasuries         | duration hedge     | 2002         |
| TLT    | 20y+ US Treasuries          | convex duration    | 2002         |
| TIP    | inflation-linked Treasuries | real-rate hedge    | 2003         |
| GLD    | gold                        | crisis diversifier | 2004         |
| ^IRX   | 13-week T-bill rate         | the cash leg       | —            |

Prices are split- and dividend-adjusted daily closes from a public quote API, converted to total-return relatives  $R_t = P_t^{\text{adj}}/P_{t-1}^{\text{adj}}$ ; the cash leg accrues the annualized T-bill rate day-counted as  $r_t^f = y_t/252$ . The exposure-rule experiments (§6, §7.4) additionally use the lab’s Dow-30 daily panel (2000–2026). Every panel enters through a signed data gate: the file hash, source URL and vintage are recorded before any experiment may read it, and panels are ground-truthed against an independent source — a discipline bought by experience (a Chinese panel once passed every internal consistency check while carrying a mixed clock; only external ground-truthing caught it).

**Point-in-time (PIT) discipline.** A decision or forecast issued at time  $t$  may use only  $\mathcal{F}_t = \sigma(\{x_s : s \leq t\})$ , with publication lags respected (e.g., inflation enters with its release delay). Evaluation “stands at” historical origins: at origin  $\tau$  the machinery sees data only up to  $\tau$ , quotes its decision or promise, and is then scored on the realized continuation — never re-fit on it.

## 4.2 Scenario kernels for the goal planner

The dynamic planner of §4.4 requires return scenarios. Futures are spliced from real history by the stationary bootstrap of Politis and Romano [1994]: blocks of consecutive daily returns with geometric length,  $\mathbb{P}(L = \ell) = p(1 - p)^{\ell-1}$ ,  $\mathbb{E}[L] = 1/p = 63$  days, resampled with shared block indices across assets (preserving cross-asset correlation). The cash leg is *not* level-resampled: interest rates are persistent, so sampled paths resample rate *changes* and cumulate from the current rate,

$$r_t^f = \max\left(0, r_0^f + \sum_{s \leq t} \Delta r_{\iota(s)}^f\right), \quad (6)$$

with  $\iota(s)$  the shared block indices — the v1.4 repair whose absence made zero-rate-era cash promises structurally overconfident (§7.3). An ensemble member re-centers paths on building-block expected returns, a stress member keeps a crash in the left tail, and a parameter-uncertainty layer widens with horizon; member weights are learned prequentially (each member weighted by errors of forecasts it locked *before* the realizations arrived). The engine’s own calibration harness is the subject of the companion paper; every probability below inherits its UNCALIBRATED RESEARCH ESTIMATE label until those gates close.

## 4.3 Evaluation metrics

For a wealth path  $W_t$  over horizon  $H$  years with goal ratio  $G$ :

$$\text{max drawdown} = \min_t \left( \frac{W_t}{\max_{s \leq t} W_s} - 1 \right), \quad \text{Sharpe} = \frac{\sqrt{252} \bar{x}}{\hat{\sigma}(x)}, \quad (7)$$

$$F_q = Q_{0.20} \left[ (W_H/W_0)^{1/H} - 1 \right], \quad (8)$$

$$\text{exceedance} = \frac{1}{|\mathcal{C}|} \sum_{(\tau, d) \in \mathcal{C}} \mathbf{1} \left[ \text{realized}_{\tau, d} \geq F_q^{(\tau, d)} \right], \quad (9)$$

$$J = \mathbb{P}(W_H \geq G) - \lambda \mathbb{E}[\max(0, G - W_H)], \quad (10)$$

where  $x_t$  in Eq. 7 is the daily excess return;  $F_q$  (Eq. 8) is the *promise* — the 20th-percentile annualized outcome, exceeded with 80% probability by construction; the exceedance (Eq. 9) is scored over the PIT cells  $\mathcal{C} = \text{origins} \times \text{capacity budgets}$ ; and  $\lambda$  in Eq. 10 is the user’s miss-magnitude weight. An 80%-confidence floor is *honest*

iff its exceedance is  $\geq 0.80$ ; Eq. 9 is therefore the flagship test — it backtests the *promises*, not the returns.

#### 4.4 The readable champion: dynamic programming on the funding ratio

Given scenario kernels, the planner solves the goal problem exactly by backward induction on (funding ratio  $\times$  time) [Das et al., 2020]. With  $V_T(W) = U(W/G)$ ,  $U(w) = \mathbf{1}[w \geq 1] - \lambda \max(0, 1 - w)$ , and a capacity-filtered menu  $\mathcal{P}$  (portfolios whose drawdown tail exceeds the user’s budget are excluded from the action set):

$$V_t(W) = \max_{p \in \mathcal{P}} \mathbb{E}[V_{t+1}(W \cdot R_{t+1}^{(p)} + c_t)], \quad (11)$$

where  $R^{(p)}$  is the scenario gross return of portfolio  $p$  and  $c_t$  scheduled contributions. The resulting policy is a *table* — one action per state — read out loud as “behind the goal  $\rightarrow$  full allowed equity; at the goal  $\rightarrow$  lock it in” [Browne, 1999]. This table is the object most of our interpretability metrics live on.

### 5 CrystalScore: measuring “legible” so it can lose

A legibility claim that cannot lose is marketing. We use

$$\text{CRYSTALSCORE} = F \times S \times St, \quad (12)$$

at a fixed parsimony budget  $K \leq 9$  (at most nine named rules/leaves), with each axis operationalized so that it *could have* come out low [Doshi-Velez and Kim, 2017, Jacovi and Goldberg, 2020, Lipton, 2018]:

$$F \text{ (faithfulness)} = \mathbb{E}[\mathbf{1}[\pi(b \leftarrow \text{do}(b^*)) = \pi^*(b^*)]], \quad (13)$$

$$S \text{ (simulatability)} = \text{acc}(\text{story}_K, \pi) \text{ s.t. } |\mathcal{R}(\text{story}_K) - \mathcal{R}(\pi)| \leq \varepsilon, \quad (14)$$

$$St \text{ (stability)} = \frac{1}{n} \sum_{i=1}^n \mathbf{1}[\text{mechanism replicates on seed } i]. \quad (15)$$

Faithfulness (Eq. 13): *do*-writes must land, judged against a reference optimum, not against the policy’s own habits. Simulatability (Eq. 14): a  $K$ -rule story must match the policy *at performance parity*  $\mathcal{R}$ . Stability (Eq. 15): the mechanism must replicate across independent training seeds. The parity constraint in Eq. 14 matters: a story that reproduces a lobotomized policy is cheating. The non-saturating companion is the **MDL deficit** [Rissanen, 1978],

$$\Delta_{\text{MDL}} = \text{Acc}(\pi) - \text{Acc}(\text{story}_K), \quad (16)$$

the accuracy lost by forcing the explanation to be short; 0 means a short story loses nothing.

**Real-data values.** On its own deployed-panel stance log, R6c scores  $F=1.00$  (latent steering does move behavior — faithfulness alone cannot distinguish the designs),  $S=0.244$ ,  $St=0.619$ : CRYSTALSCORE=0.151. CRYSTAL-1’s axes on the real US machinery:  $S=0.92$  (an 8-leaf tree reproduces the DP champion’s policy table, §4.4);  $St=1.00$  (the certified rule’s behavior replicates on 10/10 training seeds);  $F=1.00$  (the certified readable rule of Eq. 4 *is* a faithful command surface: writing the bear belief past the threshold changes exposure exactly as named);  $\Delta_{\text{MDL}}=0.0$ :

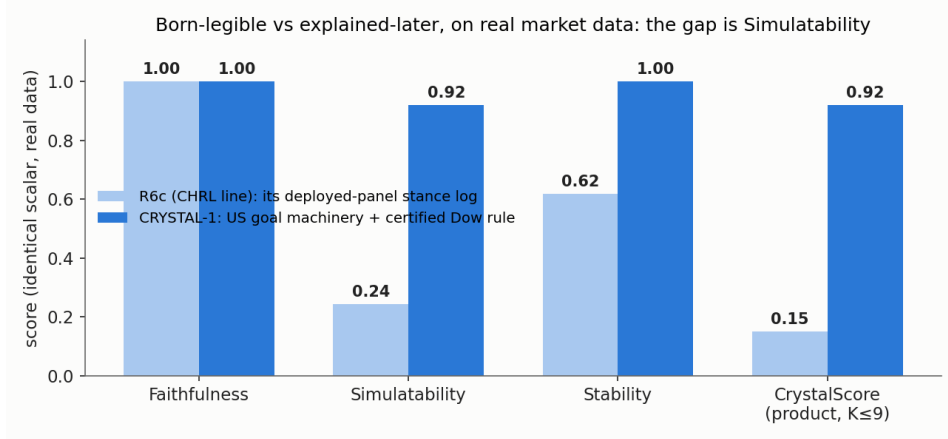


Figure 2: The identical scalar on both architectures, real data only: R6c is scored on its own deployed-panel stance log; CRYSTAL-1 on the US goal machinery (tree-vs-champion simulatability) and the certified Dow rule (10-seed stability, write fidelity). Faithfulness is 1.0 for both — steering works on a black box too; the discriminating axis is whether a  $K \leq 9$  named story can reproduce behavior at parity. Alpha does not enter the score.

CRYSTALSCORE=0.92 (Figure 2). The entire  $6\times$  gap is Simulatability — exactly what “born legible” buys. This score is for the transparent *champion*; the RL reward-dial *prototype* of §6 is a different object and scores CRYSTALSCORE=0.03 ( $F=0.03$ ) — it is legible and stable but *not* a faithful command surface, which is one more reason DP remains champion.

## 6 The behavioral-influence toolbox: four levers, each with an equation and a measured outcome

A legible agent is only useful if you can *change* it on purpose. We exercised four families of influence on CRYSTAL-1’s *PPO reward-dial heads* — a research prototype, not the champion — all on the real US daily panel (train 2010–18, held-out evaluation after), and each with a *control battery* so we report what the levers actually do, not what they appear to (Figure 3).

**R — reward shaping.** The risk-targeted reward

$$r_t = e_t x_t - \lambda_R \max(0, DD_t - D^*), \quad (17)$$

( $e_t$  exposure,  $x_t$  the asset return,  $DD_t$  running drawdown,  $D^*$  the budget) moves the exposure dial, but — stated honestly — the reward penalty *alone does not enforce the budget*: only the 5% head holds its budget (−4.81% held-out max drawdown, by parking in cash), while the 8% and 12% heads *breach* theirs (−15.9%). A drawdown matched twin shows  $z = 0.00$  timing skill: reward shaping buys *risk placement*, not timing, and cold PPO re-derives what the goal machinery already knew — the dial beats timing. Enforcing the budget requires a *separate cost objective*: the companion Lagrangian constrained head drives the interim-drawdown cost down (dual multiplier  $0 \rightarrow \sim 5$ , −0.18 breach-years vs  $\lambda=0$ ) but trades 15–37 pp of goal probability — another reason the transparent DP champion, which respects capacity by a hard pre-filter, is not displaced.

**T — teachers.** The cautionary tale first: trained cold, the heads’ argmax behavior looked sensible while the underlying preferences were nearly flat — an effectively untrained network masked by deterministic evaluation (mean max action-probability  $0.21 \approx 1/5$ ; for a behavioral scientist: inferring function from the argmax of a flat tuning curve). The cure is a teacher: behavior cloning from the certified readable rule (Eq. 4),

$$L_{\text{BC}}(\theta) = -\mathbb{E}_{(s,a^*)}[\log \pi_{\theta}(a^* | s)], \quad (18)$$

87% agreement, used as a warm start — which *resolves the collapse*: mean max action-probability  $0.21 \rightarrow 0.73$ , and the belief becomes load-bearing. The same mechanism closed the soft-tree parity gap (cold  $-2.57 \rightarrow$  annealed  $-0.27 \rightarrow$  BC-warm-started  $-0.018$ , paired  $p \approx 0.84$ ) — imitation as initialization, optimization as refinement [Ross et al., 2011, Rusu et al., 2016]. (Leakage note: the E-15 teacher was *selected* on the 2022–23 hold, so its hold read is contaminated; the leak-free teacher in the pipeline is the DP champion, and a fresh read is pre-registered for 2027.)

**I — interventions, under controls.** This is where the controls earn their keep, and the honest result overturns a first-pass claim. Belief writes (Eq. 5) and a per-primitive promotion of the *defensive leaf* appear to steer behavior — but the control battery shows that on these reward-dial heads they are *not* a clean causal command surface: belief-write fidelity is only 3%, the confident teacher-warm head is a *no-op* under a modest per-primitive bias, and on the cold near-uniform head a *matched-random* leaf moves behavior exactly as much as the defensive one (the placebo fails). The apparent “intervention works” of a first pass was a near-uniform / global-bias artifact — exactly the failure mode the controls exist to catch. One genuine positive survives: the certification gate correctly *refuses* a +1-nat promotion that fails the non-inferiority test,

$$\text{accept the write} \iff \hat{J}_{\text{post}} \geq \hat{J}_{\text{pre}} - \delta_{\text{NI}}, \quad (19)$$

and the reward “winner” that beat base on development is likewise *rejected* on the hold by the NI bar ( $z = 1.34 < 2.15$ ). Moving a deterministic output is not the same as a certified, causal control — and the reward-dial head is not the faithful belief-command surface that the champion’s readable rule (Eq. 4, faithfulness 1.0 in §5) is.

**A — architecture, and the head’s CrystalScore.** Tree depth is a legibility dial set *before* training: depth-2 yields a 4-leaf policy, depth-3 the 8-leaf one. Scoring the head itself (the audit found this was never done): it is perfectly *simulatable* (a 4-leaf story distills the 8-leaf head exactly,  $S = 1.0$ ) and *stable* across 3 seeds ( $St = 1.0$ ), but its *faithfulness* is near zero ( $F = 0.03$ ) — so its CRYSTALSCORE = 0.03. The reward-dial head is a legible, stable *dial*, not a faithful belief-driven command surface; that surface lives in the DP champion, not the RL prototype.

## 7 Real-data results

### 7.1 The readable champion beats the best static portfolio

On the flagship goal task (double the wealth in ten years, US universe, Eq. 11), the dynamic readable policy beats the *best static* portfolio by +10.5 pp goal probability (CI90 [+9.3, +11.7], stable across folds; Figure 4). An 8-leaf tree reproduces the champion’s table at 0.92 (Eq. 14) — a human can simulate it. A dedicated gate checks

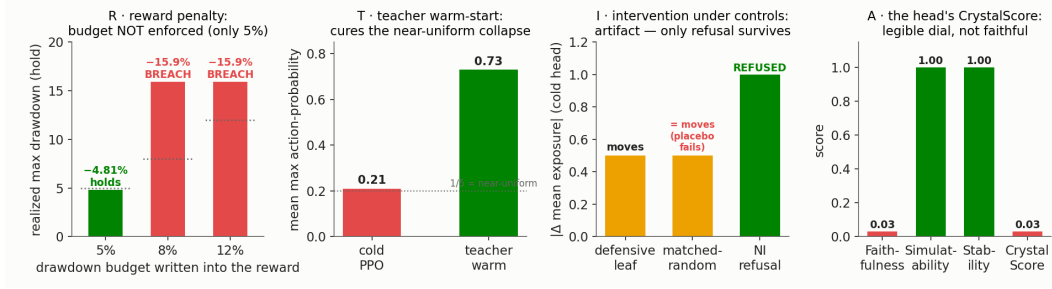


Figure 3: The four levers on the real daily panel, under controls. R (Eq. 17): the reward penalty holds only the 5% budget ( $-4.81\%$ ); the 8%/12% heads breach ( $-15.9\%$ ) — the penalty does not enforce the budget (the Lagrangian does, at a goal cost). T (Eq. 18): the teacher warm-start cures the cold near-uniform collapse (mean max action-prob  $0.21 \rightarrow 0.73$ ). I (Eqs. 5, 19): the control battery shows the intervention is NOT a clean causal control — a matched-random leaf moves the cold head as much as the defensive one (placebo fails); belief-write fidelity 3%; only the NI *refusal* survives. A: the head scores CRYSTALSCORE = 0.03 ( $F\ 0.03 \times S\ 1.0 \times St\ 1.0$ ) — a legible, stable dial, not a faithful command surface.

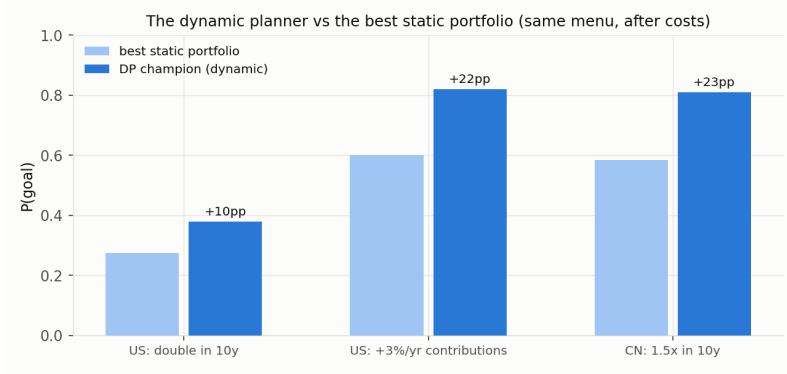


Figure 4: The readable dynamic policy vs the best static portfolio on the same menu, after costs (US universe; the CN panel shown for the separate-verdict discipline). The lift is +10.5 pp goal probability on the flagship task.

that the lift is not bought with deeper misses; on Chinese data it correctly caught exactly that (the digital-option behavior of goal-reaching policies), which moved  $\lambda$  into the user contract — the tension between goal probability and miss depth is structural, so the *user* must own it. A conditional PPO challenger (behavior-cloned from the DP teacher, hard-shielded on capacity [Achiam et al., 2017, Schaul et al., 2015]) currently *loses* to exact DP on the same action space (regret 6.6 pp); we report this because the field’s default is the opposite claim.

## 7.2 The improvement loop, profile by profile: 1% $\rightarrow$ 52% at unchanged risk

The outer layer (the subject of the companion paper) is a loop in which an LLM coding agent proposes typed machinery changes that must survive a control battery — matched-random (placebo), wrong-direction, half-dose, out-of-time — plus a one-shot held-out read. Here it matters as CRYSTAL-1’s *real-data adaptation record*: 65 iterative cycles across five investor profiles on the US universe, acceptance on the user’s declared objective  $J$  (Eq. 10), every accept immediately re-read on a held-out

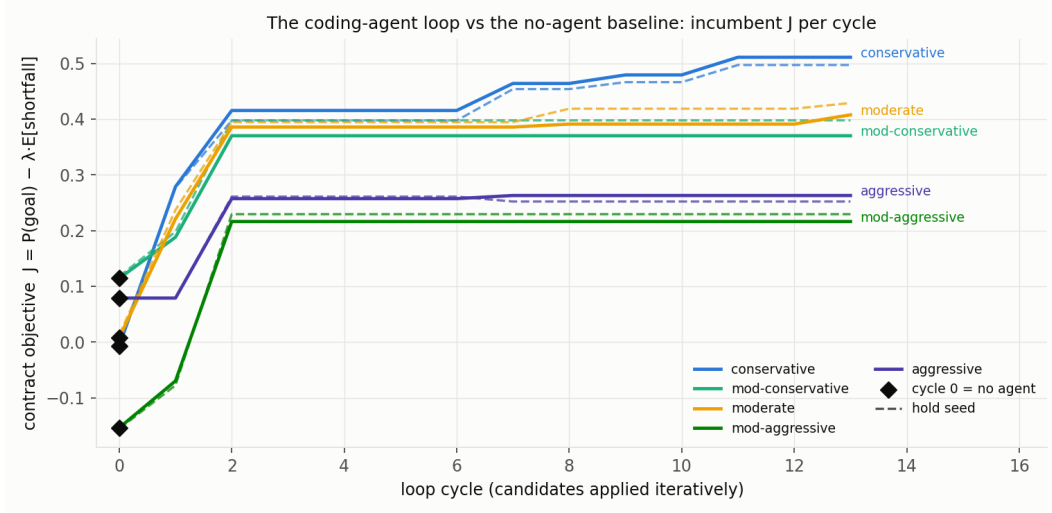


Figure 5: Learning curves on real US data: the contract objective  $J$  (Eq. 10) vs loop cycle, per profile. The leftmost point (cycle 0) is the no-agent baseline; solid = development seed, dashed = held-out. Plateaus are honest rejections; steps are accepted candidates.

seed.

| profile          | accepts/13 | $J$ : cycle 0 | $J$ : final (hold) | $\mathbb{P}(\text{goal}): 0 \rightarrow \text{final}$ |
|------------------|------------|---------------|--------------------|-------------------------------------------------------|
| conservative     | 5          | -0.007        | +0.497             | 0.01 $\rightarrow$ 0.52                               |
| mod-conservative | 2          | +0.114        | +0.398             | 0.18 $\rightarrow$ 0.46                               |
| moderate         | 4          | +0.008        | +0.429             | 0.06 $\rightarrow$ 0.48                               |
| mod-aggressive   | 2          | -0.154        | +0.229             | 0.00 $\rightarrow$ 0.35                               |
| aggressive       | 2          | +0.079        | +0.252             | 0.08 $\rightarrow$ 0.25                               |

Table 1: The per-profile improvement loop on real US data: 15 of 65 candidates accepted, all 15 confirmed on held-out data (dev  $\approx$  hold; no development-seed overfitting visible). The largest gains land exactly where the capacity filter bites hardest: the conservative profile’s goal probability rises from 1% to 52% at an *unchanged* risk budget. All numbers are research-tier scenario estimates under §4.2.

### 7.3 The promise backtest: do the quoted floors come true?

For each profile we sweep the capacity budget across its band and, at every point-in-time origin (semi-annual from 2010), quote the promise  $F_q$  (Eq. 8), then run the *same policy* on the realized subsequent years and score Eq. 9. The first run failed the conservative profile at 48.5% — and the failure was the finding: the promise there is cash-dominated, and level-resampled rates let pre-crisis 4–5% yields leak into zero-rate-era promises. The persistent-rate repair (Eq. 6) lifted exactly the diagnosed cells: conservative exceedance 48.5%  $\rightarrow$  84.2%, and **all five US profiles now clear the 80% bar** across 576 PIT cells; the risk (drawdown) promises held in 100% of all 936 cells on both markets (Figure 6). China — evaluated separately, never averaged — passes 1/5 with three diagnosed mechanisms and named fixes queued. *Evidence tier, stated plainly:* the v1.4 pass re-uses the windows whose failures motivated the fix — development-tier by our own taxonomy; the genuine forward test is pre-registered (first scoring read 2027-07-12; an untouched multi-year fold matures 2029).

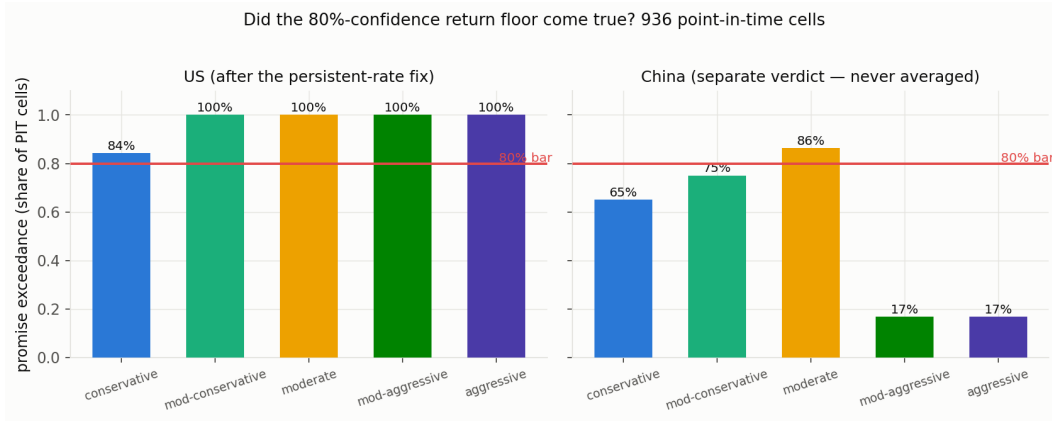


Figure 6: Promise exceedance (Eq. 9) per profile against the 80% bar. Left: US after the persistent-rate fix — 5/5 profiles pass. Right: China, reported separately; three failures with diagnosed mechanisms. Risk promises: 100% of all 936 cells.

#### 7.4 A certified readable rule on untouched data

The loop’s frozen gate has certified exactly one real-data policy to date — and it is a sentence (Eq. 4). It entered with 7 accepts and 0 placebo accepts, and held one-shot on 629 untouched trading days: Sharpe 1.46 vs 1.39 buy-and-hold, max drawdown  $-12.9\%$  vs  $-15.5\%$ , risk-adjusted improvement  $z = +2.06$ . The claim is a *risk* claim (same return, meaningfully less pain), certified as such; a written return-improvement rule was separately tested and *closed* ( $-8.8\%$ ). A fully autonomous lifetime run then refined this rule under the same gates: 40 proposals  $\rightarrow$  4 certified refinements, with the evidence budget self-sustaining ( $0.20 \rightarrow 0.40$ ).

#### 7.5 The transparency audit

The certified rule’s machinery was audited on the real panel: naming faithfulness 1.0 (the state called “bear” is the state in which de-risking is optimal under the world model); 10-seed behavioral stability 1.0; rule re-discovery from scratch 3/3; belief MDL deficit 0.0 (Eq. 16). These are the CRYSTAL-1 entries of Figure 2 and Table 2. Their honest scope: measured on one substrate and one rule family — hardening them across substrates, seeds and profiles is Month 1 of the research program (§11).

Joseph: H1 asks whether the machine-simulatability 0.92 predicts *human* simulatability — your psychophysics.

## 8 The designed-market experiment: can a market reward only a clean mind?

Real markets have very noisy logic: the regime is largely priced, so the honest value of even a perfect regime belief can be zero — and a null result on real data cannot distinguish “the instrument is broken” from “there was nothing to detect.” This motivates one deliberately synthetic experiment, asked as a question: *can one build a market that rewards only a “clean” mind — an environment where the belief information carries value by construction, so that command-surface machinery can be calibrated at all?*

We built such an environment (a regime market whose returns are generated from latent states the belief can, in principle, track) and used it strictly as **instrument calibration** — none of the following is market-performance evidence. On it:

|                           | CHRL / R6c                                                | CRYSTAL-1                                                                                                                |
|---------------------------|-----------------------------------------------------------|--------------------------------------------------------------------------------------------------------------------------|
| Memory                    | 64-d unnamed latent, smeared; blast radius unboundable    | named posterior $b_t$ (Eq. 2); the sole memory channel                                                                   |
| Policy legibility         | post-hoc codes over a continuous latent; story acc. 0.244 | the policy <i>is</i> a table/rule; 8-leaf story acc. 0.92 at parity                                                      |
| Steering primitive        | force the latent toward a K-means centroid (correlation)  | named <i>do-write</i> (Eq. 5), fidelity 1.0                                                                              |
| Refusals                  | OOD gate shrinks or zeroes steps                          | the NI bar (Eq. 19) refused a +1-nat promotion                                                                           |
| Control surface           | ~214 knobs                                                | 5 named levers                                                                                                           |
| Seed stability            | 0.619                                                     | 1.00 (10 seeds)                                                                                                          |
| CRYSTALSCORE (Eq. 12)     | <b>0.151</b>                                              | <b>0.92</b>                                                                                                              |
| Flagship real-data result | walk-forward validation: Sharpe 1.09, maxDD $-7.34\%$     | certified rule, 629 untouched days: Sharpe 1.46 vs 1.39, maxDD $-12.9\%$ vs $-15.5\%$ ; DP champion +10.5 pp             |
| Deployment status         | deployed: live firewall, operating history                | the goal machinery live on the US universe (paper track); client certification still <i>refused</i> , with dated unlocks |

Table 2: The final comparison, real data only. The honest split survives in refined form: R6c remains the longer-operating deployed system; CRYSTAL-1 wins every legibility and control axis *and* now carries its own real-data results, while its client-facing certification correctly refuses until the pre-registered reads mature.

belief writes were *proved* causal against an exogenous belief-MDP optimum (agreement 0.80 corner head / 0.835 tree head); a likelihood-based *lie detector* separates sharp-but-wrong commands from uncertain-but-right ones (excess predictive NLL 0.43 vs  $\approx 0$ ; detection 0.71 at 5% false alarms), while the naïve entropy gate points *backwards* (it trusts the confident lie); a 216-cell factorial showed command composition is non-additive with *sign-flipping* cross-terms (median  $|\varepsilon| = 2.66$ , both signs observed, surviving raw logits) — which is why the write surface is governed by a calibrated ledger,  $D = \sum |\alpha_i| \tau_i + \sum |\varepsilon_{ij}| \tau$ , a co-activation cap  $N_{\text{live}} \leq K$ , a kNN manifold governor (24/24 engineered off-manifold corners flagged at 1.2% held-out FPR) and a writ ladder whose own false-accept rate has a first estimate (0/48;  $\lesssim 6\%$  by the rule of three). On a *hard* designed substrate the gate also certified a genuine return $\leftrightarrow$ legibility *tension* — legibility is not free everywhere, and anyone claiming otherwise is measuring easy tasks.

The bridge back to reality is an explicit **value-of-information gate** (Figure 7): the designed market shows VoI = 5.92 (+282% of a belief-blind baseline) and the gate *opens*; on a real book the measured VoI is 0 and the gate *closes* — so nothing calibrated on the designed market is allowed to generate a real-market claim. The one real-panel port of this command surface (a belief-driven exposure mode) exists behind that gate and an enable table, and is deliberately *not* cited as evidence anywhere in §7.

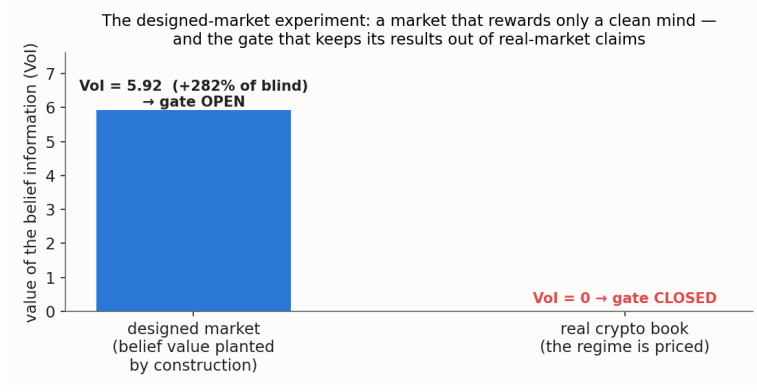


Figure 7: The designed-market experiment and its fence. Left: the environment built to reward a clean mind — the belief information is worth +282% of blind by construction, so command machinery (causal writes, the lie detector, composition governance) can be calibrated. Right: the same measurement on a real book returns zero, and the gate closes — designed-market results are instrument calibration, never market evidence.

## 9 What did not work, and what bounds the claims

A paper about honest machinery should list its own falsifications; each cost us a believed hypothesis, and the machinery that killed them is the contribution. (i) A famous valuation indicator (**CAPE**): made real-data forecasts worse → falsified, reverted. (ii) The **PPO challenger** loses to the readable table on real data (regret 6.6 pp) → reported, not hidden. (iii) A **Nasdaq-tracking sleeve** dominated every Chinese profile → red-flagged *by the agent itself*: its edge is one untestable sample era. (iv) The **conservative promise failed its first backtest** (48.5% vs 80%) → the diagnosis became Eq. 6 and repaired exactly the diagnosed cells (→ 84.2%). (v) A **written return-improvement rule** was tested and closed (−8.8%) — the certified claim is risk-shaped, not alpha-shaped. (vi) On a hard *designed* substrate, **legibility is provably not free** (the certified tension of §8) — the open frontier of the interpretability track. (vii) Post-hoc interpretability on CHRL **was not worthless**: it found the load-bearing concepts (motifs, codes, counterfactual replay) that CRYSTAL-1 promoted from probes into architecture — the right summary is that building them in made the story  $\sim 4\times$  more compressible (0.244 → 0.92 at the same budget).

## 10 Related work

Interpretability-by-construction vs post-hoc explanation [Rudin, 2019, Lipton, 2018]; rigor and evaluation of interpretability [Doshi-Velez and Kim, 2017, Hoffman et al., 2018]; faithfulness as a criterion [Jacovi and Goldberg, 2020]. Belief-state control [Kaelbling et al., 1998] and regime models [Hamilton, 1989] supply the named memory; soft trees [Frosst and Hinton, 2017] the legible head; MDL [Rissanen, 1978] the non-saturating ruler; interventional causality [Pearl, 2009] the faithfulness protocol. Discretized latent-action behavior generation [Lee et al., 2024] pursues the complementary direction — learned discrete codes for *expressiveness* — where CrystalRL fixes the codes to *named* coordinates for legibility; goal-based planning [Das et al., 2020, Browne, 1999]; hierarchical RL as partial legibility [Sutton et al., 1999, Bacon et al., 2017]; PPO and constrained variants [Schulman et al., 2017, Achiam et al.,

2017]; goal-conditioned value functions [Schaul et al., 2015]; imitation-then-optimize [Ross et al., 2011, Rusu et al., 2016]; reward shaping [Ng et al., 1999]; the stationary bootstrap [Politis and Romano, 1994]. The self-improvement layer — LLM proposers over verified evaluators [Romera-Paredes et al., 2024, Novikov et al., 2025, Ma et al., 2024, Wang et al., 2023] and the statistics of adaptive search over noisy financial evaluators [Bailey and López de Prado, 2014] — is treated in the companion paper (*Hello Crystal*); this paper builds the object those loops are allowed to modify.

## 11 The research program: twelve hypotheses, each with a kill test

This section is deliberately ambitious — the point of a legible, steerable, self-modifying agent is not one trading policy but a *general method for keeping evolving systems readable*. Each hypothesis is falsifiable; each names the observation that kills it.

1. **H1 — Machine simulatability predicts human simulatability.** The 0.92 story accuracy is machine-measured. *Test:* psychophysics — humans predict the policy’s action in unseen cells of the real DP tables. *Kill:* human accuracy at chance while machine simulatability stays at ceiling.
2. **H2 — A legibility scaling law.** Optimal vocabulary size  $K$  tracks substrate behavioral complexity across real universes — a “Kolmogorov frontier” for policies. *Kill:* substrates where optimal  $K$  is unrelated to measured complexity.
3. **H3 — The return $\leftrightarrow$ legibility tension is a substrate property, not an architecture property.** Prediction: the tension is maximal where the regime is fully priced and vanishes where belief information carries value. *Kill:* tension persists on a designed  $\text{VoI} > 0$  environment.
4. **H4 — Interpretability drift.** Under the self-improvement loop, naming faithfulness decays unless actively re-certified (semantic drift of the belief coordinates). *Test:* track Eq. 13 across accepted loop generations on the real machinery. *Kill:* faithfulness stays at 1.0 over  $\geq 20$  accepted changes with no re-certification.
5. **H5 — Motif conservation across architectures.** CHRL’s discovered primitives re-emerge as CRYSTAL-1 rule clauses (homologous behaviors across “species”). *Test:* match the frozen-test primitives to leaf semantics on the same real windows. *Kill:* no better-than-chance correspondence.
6. **H6 — The write protocol is pharmacological.** SET\_BELIEF dose-response on the real panel is monotone and saturating. *Kill:* non-monotone dose regions on any certified substrate.
7. **H7 — Composition governance is necessary on real data.** Sign-flipping command interactions (so far shown only on the designed market, off-manifold) appear on the real panel’s visited manifold. *Kill (of the premise):* on-manifold real-data interactions are additive — then the ledger’s  $\varepsilon$  term is precautionary overhead.
8. **H8 — The lie detector generalizes.** Authority-tracks-NLL discriminates wrong-but-sharp commands on real panels at operationally useful rates. *Kill:* detection at chance outside the designed market.

GAP: position vs concept-bottleneck models (Koh et al. 2020) and vs circuit-style transformer interpretability — one paragraph each, Joseph’s lit pass.

Joseph: this program is yours to reshape: pick one hypothesis per month, or replace them with better ones — the kill-test discipline is the only non-negotiable.

9. **H9 —  $\alpha_{\text{machine}}$ , full-firewall.** The 0/48 core proxy holds ( $\leq 5\%$ ) when estimated through the complete real-data gate stack. *Kill*: full-stack estimate an order worse.
10. **H10 — Faithful client-facing explanations.** Every “why” sentence shown to a user can be made *measurably* faithful: perturb the stated reason, behavior must move accordingly. *Kill*: explanations that pass fluency review but fail perturbation faithfulness at scale.
11. **H11 — Certified vocabulary growth.** GROW\_K can be run as a certified architectural operation: the agent learns a new regime word without semantic drift of the old ones. *Kill*: any old-coordinate faithfulness drop after growth.
12. **H12 — The score applies to the improver.** CrystalScore’s axes transfer to the *coding agent’s* own proposals — interpretability of the improver, not just the improved. *Test*: score proposal rationales against their measured effects. *Kill*: rationale–effect agreement at chance (the LLM’s stated reasons are confabulation — itself a publishable result).

## 12 Limitations

(i) The scenario-based probabilities of §7 remain UNCALIBRATED RESEARCH ESTIMATES; the calibration gates (7/10 green) and their dated forward reads (2027-07-12; 2029) live in the companion paper. (ii) The v1.4 promise-backtest pass re-uses the windows whose failures motivated the fix — development-tier evidence. (iii) The real-data CrystalScore’s simulatability axis is measured on the DP champion and the certified rule family — not yet on every head (Month 1 of the program). (iv) The command-surface *proofs* (causal writes vs an exogenous optimum, the lie detector, composition governance) are designed-market calibrations (§8); their real-data analogues are H6–H9. (v) China passes 1/5 profiles; its named fixes are queued, and no CN number is averaged into any US claim. (vi) Execution economics beyond 10 bp switching costs (taxes, spreads, gaps) are only partially bounded.

## Acknowledgments and drafting discipline

Prepared *write-paper-first* [Eisner, 2010]: every margin `todo` marks a real gap. Experiments were run and logged under the repository’s honesty contract — every entry names its null, its seed, and its worst caveat — in the spirit of Feynman [1974]: “*The first principle is that you must not fool yourself — and you are the easiest person to fool.*” LLM coding agents (Claude, Codex) operated the loops under the stated controls; the agents propose, the gates dispose.

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